

The Greebling

Scaling Neural-Data Vaults and AI Urbanization in the
Bitsmasher Network

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Infrastructure expansion within the Skynet AI Testing Field necessitates a transition from legacy flat-file storage to high-density SQL architectures. This session details the technical recovery and scaling of the blowfish node, focusing on the 274GB filesystem exhaustion event, OpenBSD disk re-initialization, and the deployment of the Hub 07 Neural-Data Vault to support the NPUSpatialEngine and T2BM pipeline.

The 274GB Log Bloat Crisis

- **Detection:** Filesystem exhaustion on the /home partition of the blowfish database host.
- **Root Cause:** Rapid expansion of `mysqld.log` reaching 255 GiB, indicative of service-level error loops.
- **Emergency Response:** Truncation of the log file and implementation of `newsyslog` rotation policies using the `rcctl restart mysqld` directive.

Storage Node Re-Initialization

- **Hardware:** Integration of Western Digital 2.8TB My Passport and 1TB Black drives.
- **Filesystem:** Utilization of GUID Partition Table (GPT) and OpenBSD Fast File System (FFS).
- **Optimization:** Alignment for 4096-byte sectors and establishing permanent mounts via DUID in `/etc/fstab` with `nodev` and `nosuid` security flags.

The Neural-Data Vault (Hub 07)

- **Database Migration:** Relocating the MariaDB `datadir` to the 1TB expansion volume to offload BlueMap SQL tiles.
- **Schema Design:** Deployment of the `skynet_vault` database and `build_history` table.
- **Telemetry Tracking:** Implementation of the `ai_hardware` field to audit inferences from the Hailo-8L NPU and Google Edge TPU.

Spatial Inference and Density Logic

- **NPUSpatialEngine:** Simulating neural signal spread using 10x10 block cells.
- **Collision Detection:** Utilizing SQL-backed build history to prevent structural overlaps in the Testing Field.
- **Vector Selection:** NPU Density Logic for optimal site selection based on 'cluster' or 'void' preferences.

- **Transport:** Configuring NFSv3 over TCP (`-3`, `-T`) for stability.
- **Troubleshooting:** Resolving `rpc.mountd` authentication failures and `unmatched host` errors on the `chonk` node.
- **Access Control:** Transitioning to `no_root_squash` to ensure consistent ownership for the `_mysql` user across the cluster.

BlueMap and High-Resolution SQL Storage

- **Transition:** Moving from legacy storage: "file" to storage: "sql" for the dreamland (overworld) map.
- **Infrastructure Impact:** Managing high-volume SQL writes generated by the hires-layer and architectural telemetry.
- **Persistence:** Ensuring the 20 TPS performance ceiling is maintained during concurrent re-render operations.

Future Directions: Urbanizing the Testing Field

- **T2BM Pipeline:** Text to Building in Minecraft via Large Language Models.
- **Deep-Rail Integration:** Expanding the transit network from Hub 01 through Hub 07.
- **Ledger Synchronization:** Continuous updates to the GitHub technical ledger to secure infrastructure state.